Solutions Tutorial 2

1. Find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ for each of the following cases:

Answer.

$f(x,y)$	$f_x = \frac{\partial f}{\partial x}$	$f_y = \frac{\partial f}{\partial y}$
(a) $x^3y + e^x$	$3x^2y + e^x$	x^3
(b) xe^{2x+3y}	$e^{3y}(2xe^{2x} + e^{2x}) = e^{2x+3y}(2x+1)$	$3xe^{2x+3y}$
(c) $y\sin(x^2+y^2)$	$2xy\cos(x^2+y^2)$	$2y^2\cos(x^2+y^2) + \sin(x^2+y^2)$
(d) $\frac{x-y}{x+y}$	$\frac{(x+y)-(x-y)}{(x+y)^2} = \frac{2y}{(x+y)^2}$	$\frac{-(x+y)-(x-y)}{(x+y)^2} = \frac{-2x}{(x+y)^2}$
(e) $\tan^{-1}\left(\frac{x}{y}\right)$	$\frac{1/y}{1+(x/y)^2} = \frac{y}{x^2+y^2}$	$\frac{-x/y^2}{1+(x/y)^2} = \frac{-x}{x^2+y^2}$
(f) $\sqrt{x^2+y^2}$	$\frac{x}{\sqrt{x^2+y^2}}$	$\frac{y}{\sqrt{x^2+y^2}}$
(g) $(1+y^2)^{\cos x}$	See below	$2y(1+y^2)^{(\cos x)-1}\cos x$
(h) $x^{\ln y}$	$(\ln y)x^{\ln y - 1}$	See below

(g) $f_x = \frac{\partial}{\partial x}(1+y^2)^{\cos x} = \ln(1+y^2)\cdot(1+y^2)^{\cos x}\,\frac{d}{dx}\cos x = -(\sin x)(1+y^2)^{\cos x}\ln(1+y^2)$

(h) $f_y = \frac{\partial}{\partial y}x^{\ln y} = \ln x\cdot x^{\ln y}\,\frac{\partial}{\partial y}\ln y = \ln x\cdot x^{\ln y}\cdot\frac{1}{y} = \frac{\ln x\cdot x^{\ln y}}{y}$.

Recall: $\frac{d}{dx}\tan^{-1}\big(f(x)\big) = \frac{1}{1+(f(x))^2}f'(x)$.

2. If $f(x,y,z) = x^2y - 2x^2z + 3xyz - y^2z + 2xz^2$, find f_x, f_y and f_z.

Solution. $f_x = 2xy - 4xz + 3yz + 2z^2$, $f_y = x^2 + 3xz - 2yz$, $f_z = -2x^2 + 3xy - y^2 + 4xz$.

3. A function of two variables $f(x,y)$ is defined as follows

$$f(x,y) = \begin{cases} \frac{2x^3y+2x^4+6y^2+5x^5}{x^4+3y^2} & (x,y) \neq (0,0) \\ 2 & (x,y) = (0,0) \end{cases}$$

Calculate $\left.\frac{\partial f}{\partial x}\right|_{(0,0)}$ and $\left.\frac{\partial f}{\partial y}\right|_{(0,0)}$ using the definition.

Solution. $\left.\frac{\partial f}{\partial x}\right|_{(0,0)} = \lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0}\frac{1}{h}\left(\frac{2h^4+5h^5}{h^4} - 2\right) = \lim_{h\to 0}\frac{2+5h-2}{h} = 5$.

$\left.\frac{\partial f}{\partial y}\right|_{(0,0)} = \lim_{h\to 0}\frac{f(0,h)-f(0,0)}{h} = \lim_{h\to 0}\frac{1}{h}\left(\frac{6h^2}{3h^2} - 2\right) = 0$.

4. Write down the equations of the tangent planes to the graphs of the following functions at the points $\big(a,b,f(a,b)\big)$.

(a) $f(x,y)=e^x\cos y$ at $(a,b)=(0,0)$ (b) $f(x,y)=\sin xy$ at $(a,b)=(1,\frac{\pi}{2})$

(c) $f(x,y)=\frac{y^2}{x^3}$ at $(a,b)=(1,-1)$ (d) $f(x,y)=e^{x+y}$ at $(a,b)=(2,-2)$

Solution. Tangent plane formula for $z=f(x,y)$ (same as linear Taylor's expansion):

$$z=f(a,b)+(x-a)f_x(a,b)+(y-b)f_y(a,b)$$

	(a) at (0, 0)	(b) at $(1,\frac{\pi}{2})$	(c) at (1, –1)	(d) at (2, –2)
$f(x,y)$	$e^x\cos y$	$\sin xy$	y^2x^{-3}	$e^{x+y}=e^xe^y$
$f(a,b)$	1	1	1	1
$f_x(x,y)$	$e^x\cos y$	$y\cos xy$	$-3y^2x^{-4}$	e^xe^y
$f_x(a,b)$	1	0	–3	1
$f_y(x,y)$	$-e^x\sin y$	$x\cos xy$	$2yx^{-3}$	e^xe^y
$f_y(a,b)$	0	0	–2	1
$f_{xx}(x,y)$	$e^x\cos y$	$-y^2\sin xy$	$12y^2x^{-5}$	e^xe^y
$f_{xx}(a,b)$	1	$-\frac{\pi^2}{4}$	12	1
$f_{xy}(x,y)$	$-e^x\sin y$	$\cos xy - xy\sin xy$	$-6yx^{-4}$	e^xe^y
$f_{xy}(a,b)$	0	$-\frac{\pi}{2}$	6	1
$f_{yy}(x,y)$	$-e^x\cos y$	$-x^2\sin xy$	$2x^{-3}$	e^xe^y
$f_{yy}(a,b)$	–1	–1	2	1

Tangent plane: (a) $z=1+x(1)+y(0)=1+x$; (b) $z=1$;

(c) $z=1-3(x-1)-2(y+1)$; (d) $z=1+(x-2)+(y+2)$.

5. Find f_{xx} and f_{yy} and verify that $f_{xy}=f_{yx}$ when

(a) $f(x,y)=x^2\sin y+y^2\cos x$ (b) $f(x,y)=\frac{y}{x}\ln x$

Solution. We verify $f_{xy}=f_{yx}$ from the following table.

	f_x	f_y	f_{xx}	f_{yy}	f_{xy}	f_{yx}
(a)	$2x\sin y$ $-y^2\sin x$	$2y\cos x$ $+x^2\cos y$	$2\sin y$ $-y^2\cos x$	$2\cos x$ $-x^2\sin y$	$2x\cos y$ $-2y\sin x$	$2x\cos y$ $-2y\sin x$
(b)	$\frac{y}{x^2}(1-\ln x)$	$\frac{\ln x}{x}$	$\frac{y}{x^3}(2\ln x-3)$	0	$\frac{1}{x^2}(1-\ln x)$	$\frac{1}{x^2}(1-\ln x)$

6. Let $f(x,y,z)=\frac{y\cos(xz)}{x^2+z^2}$, $x,z\neq 0$. Find f_{xyzy}.

Solution. By interchanging the order of differentiation,

$$f_y=\frac{\cos(xz)}{x^2+z^2}\Rightarrow f_{yy}=0\Rightarrow f_{xyzy}=f_{yyxz}=0.$$

7. Find the gradient of the functions (a) to (d) in Q1 at the point

(a) (1, 1) (b) (1, 0) (c) $(\sqrt{\pi},0)$ (d) $(\frac{1}{2},0)$.

Answer. (a) $f_x(1,1)=3+e$, $f_y(1,1)=1$, $\nabla f=(3+e,1)$.

(b) $f_x(1,0)=3e^2$, $f_y(1,0)=3e^2$, $\nabla f=(3e^2,3e^2)$.

(c) $f_x(\sqrt{\pi},0)=0$, $f_y(\sqrt{\pi},0)=0$, $\nabla f=(0,0)$.

(d) $f_x(\frac{1}{2},0)=0$, $f_y(\frac{1}{2},0)=-4$, $\nabla f=(0,-4)$.

$f(x,y)$	$f_x=\frac{\partial f}{\partial x}$	$f_y=\frac{\partial f}{\partial y}$
(a) x^3y+e^x	$3x^2y+e^x$	x^3
(b) xe^{2x+3y}	$e^{3y}(2xe^{2x}+e^{2x})=e^{2x+3y}(2x+1)$	$3xe^{2x+3y}$
(c) $y\sin(x^2+y^2)$	$2xy\cos(x^2+y^2)$	$2y^2\cos(x^2+y^2)+\sin(x^2+y^2)$
(d) $\frac{x-y}{x+y}$	$\frac{(x+y)-(x-y)}{(x+y)^2}=\frac{2y}{(x+y)^2}$	$\frac{-(x+y)-(x-y)}{(x+y)^2}=\frac{-2x}{(x+y)^2}$

Extra Exercises

8. Given $f(x,y)=x^2y^3$. Find f_{xx}, f_{xyy}, f_{yxy}.

Solution. $f_x = 2xy^3, f_{xx} = 2y^3, f_{xy} = 6xy^2, f_{xyy} = f_{yxy} = 12xy$.

9. Let $f(x,y)=\begin{cases}\frac{x^2y}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$. Compute, directly from the definitions:

(a) $\frac{\partial f}{\partial x}(0,0)$ (b) $\frac{\partial f}{\partial y}(0,0)$ (c) $\frac{d}{dt}f(t,t)\Big|_{t=0}$

Solution. (a) $\frac{\partial f}{\partial x}(0,0)=\lim_{h\to0}\frac{f(0+h,0)-f(0,0)}{h}=\lim_{h\to0}\frac{0}{h}=0$.

(b) $\frac{\partial f}{\partial y}(0,0)=\lim_{h\to0}\frac{f(0,h)-f(0,0)}{h}=\lim_{h\to0}\frac{0}{h}=0$.

(c) Note that $f(t)=f(t,t)=\begin{cases}\frac{t^3}{t^2+t^2}, & t\neq0\\ 0, & t=0\end{cases}=\begin{cases}\frac{1}{2}t, t\neq0\\ 0, t=0\end{cases}=\frac{1}{2}t$.

So, $\frac{d}{dt}f(t,t)\Big|_{t=0}=\frac{d}{dt}\frac{t}{2}\Big|_{t=0}=\lim_{h\to0}\frac{\frac{1}{2}(t+h)-\frac{1}{2}t}{h}\Big|_{t=0}=\lim_{h\to0}\frac{1}{2}\Big|_{t=0}=\frac{1}{2}\Big|_{t=0}=\frac{1}{2}$.

10. Find all first partial derivatives and evaluate at the given point:

(a) $f(x,y,z)=x^3y^4z^5$ at $(0,-1,-1)$. (b) $w(x,y,z)=\ln(1+e^{xyz})$ at $(2,0,-1)$.

(c) $g(x,y)=\frac{1}{\sqrt{x^2+y^2}}$ at $(-3,4)$

Solution. (a) $f_x(x,y,z)=3x^2y^4z^5 \Rightarrow f_x(0,-1,-1)=0$
$f_y(x,y,z)=4x^3y^3z^5 \Rightarrow f_y(0,-1,-1)=0$
$f_z(x,y,z)=5x^3y^4z^4 \Rightarrow f_z(0,-1,-1)=0$

(b) $w_x(x,y,z)=\frac{yze^{xyz}}{1+e^{xyz}} \Rightarrow w_x(2,0,-1)=0$
$w_y(x,y,z)=\frac{xze^{xyz}}{1+e^{xyz}} \Rightarrow w_y(2,0,-1)=-1$
$w_z(x,y,z)=\frac{xye^{xyz}}{1+e^{xyz}} \Rightarrow w_z(2,0,-1)=0$

(c) $g_x(x,y)=-\frac{x}{(x^2+y^2)^{3/2}} \Rightarrow g_x(-3,4)=\frac{3}{125}$
$g_y(x,y)=-\frac{y}{(x^2+y^2)^{3/2}} \Rightarrow g_y(-3,4)=-\frac{4}{125}$

Note that $g_x(x,y)=g_y(y,x)$ because $g(x,y)=g(y,x)$.

Check that $g_y(4,-3)=\frac{3}{125}=g_x(-3,4)$ and $g_x(4,-3)=-\frac{4}{125}=g_y(-3,4)$.

11. Show that the function $z(x,y)=\frac{x+y}{x-y}$ satisfies $x\frac{\partial z}{\partial x}+y\frac{\partial z}{\partial y}=0$.

Solution. $z(x,y)=\frac{x+y}{x-y}\Rightarrow z_x=\frac{-2y}{(x-y)^2}, z_y=\frac{2x}{(x-y)^2}\Rightarrow xz_x+yz_y=0$.

Alternative. Write $(x-y)z=x+y$.

Take $\frac{\partial}{\partial x}$: $(x-y)z_x+z=1\underset{\times y}{\Rightarrow}x(x-y)z_x+xz=x \quad (1)$

Take $\frac{\partial}{\partial y}$: $(x-y)z_y-z=1\underset{\times y}{\Rightarrow}y(x-y)z_x-yz=y \quad (2)$

$(1)+(2)$: $(x-y)(xz_x+yz_y)+(x-y)z=x+y$

$\Rightarrow xz_x+yz_y+z=\frac{x+y}{x-y}=z\Rightarrow xz_x+yz_y=0$

12. A surface $z(x,y)$ is defined by $zy-y+x=\ln(xyz)$.

(a) Compute z_x and z_y in terms of x, y, z.

(b) Evaluate z_x and z_y at $\left(x,y,z\right)=\left(-1,-2,\frac{1}{2}\right)$.

Solution. Note that $x,y,z\neq 0$.

(a) Take $\frac{\partial}{\partial x}$: $yz_x+1=\frac{yz+xyz_x}{xyz}=\frac{1}{x}+\frac{1}{z}z_x\Rightarrow z_x=\frac{z(1-x)}{x(yz-1)}, yz\neq 1$, and if $yz=1$, then $1-\frac{1}{z}+x=\ln(x)$ *. Thus, $-z^{-2}z_x+1=\frac{1}{x}\Rightarrow z_x=z^2\left(1-\frac{1}{x}\right)=\frac{1}{y^2}\left(1-\frac{1}{x}\right)=\frac{x-1}{xy^2}$ **. But $z=\frac{1}{y}$ then $z_x=0$, so $x=1$ from ** and $z=1$ from *. So $y=1$ from $yz=1$. But $x=y=z=1$ contradicts to $zy-y+x=\ln(xyz)$. Now take $\frac{\partial}{\partial y}$:

$$yz_y+z-1=\frac{xz+xyz_y}{xyz}=\frac{1}{y}+\frac{1}{z}z_y\Rightarrow z_y=\frac{z(1+y-yz)}{y(yz-1)},\ yz\neq 1.$$

If $yz=1$, then $yz_y+z-1=\frac{xz+xyz_y}{xyz}=z+yz_y\Rightarrow -1=0$, a contradiction.

(b) $z_x\left(-1,-2,\frac{1}{2}\right)=\frac{1}{2}$ and $z_y\left(-1,-2,\frac{1}{2}\right)=0$.

13. Let $f(x,y)=\begin{cases}\frac{x^2-2y^2}{x-y}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$. Use the definition to find $f_x(0,0)$ and $f_y(0,0)$.

Solution. $f_x(0,0)=\lim\limits_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=\lim\limits_{h\to 0}\frac{h}{h}=1$ and $f_y(0,0)=\lim\limits_{h\to 0}\frac{f(0,h)-f(0,0)}{h}=\lim\limits_{h\to 0}\frac{2h}{h}=2$.

14. Define the function $f(x,y)=\begin{cases}\frac{(x+2y)^2}{x+y}, & x+y\neq 0\\ 0, & x+y=0\end{cases}$.

(a) Evaluate, if possible, $\frac{\partial f}{\partial x}(0,0)$ and $\frac{\partial f}{\partial y}(0,0)$.

(b) Is $f(x,y)$ continuous at $(0,0)$?

Solution. (a) $\frac{\partial f}{\partial x}(0,0)=\lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=\lim_{h\to 0}\frac{\frac{h^2}{h}-0}{h}=1$ and

$$\frac{\partial f}{\partial y}(0,0)=\lim_{h\to 0}\frac{f(0,h)-f(0,0)}{h}=\lim_{h\to 0}\frac{\frac{4h^2}{h}}{h}=4.$$

(b) Consider pathways $x=-2y$ and $x=-y+y^2$,

$$\lim_{(x,y)\to(0,0)}f(x,y)=\lim_{y\to 0}\frac{0}{-2y+y}=0 \text{ and}$$

$$\lim_{(x,y)\to(0,0)}f(x,y)=\lim_{y\to 0}\frac{(-y+y^2+2y)^2}{-y+y^2+y}=\lim_{y\to 0}\frac{(y^2+y)^2}{y^2}=\lim_{y\to 0}(y+1)^2=1.$$

Hence, $f(x,y)$ is not continuous at $(0,0)$.

15. Is it reasonable to say that the surfaces of $x^2+y^2+(z-1)^2=1$ and $x^2+y^2+(z+1)^2=1$ are tangent to each other at (0, 0, 0)?

Solution. Tangent plane's equation is given by $\nabla f(x_0,y_0,z_0)\cdot\left(x-x_0,y-y_0,z-z_0\right)=0$.

Given that $x_0=y_0=z_0=0$, so the equation is $xf_x(0,0,0)+yf_y(0,0,0)+zf_z(0,0,0)=0$.

Let $F(x,y,z)=x^2+y^2+(z-1)^2-1$ and $G(x,y,z)=x^2+y^2+(z+1)^2-1$. Now

$$\left(F_x,F_y,F_z\right)=\left(2x,2y,2(z-1)\right)\Rightarrow\left(F_x,F_y,F_z\right)\Big|_{(0,0,0)}=\left(0,0,-2\right)$$
$$\left(G_x,G_y,G_z\right)=\left(2x,2y,2(z+1)\right)\Rightarrow\left(G_x,G_y,G_z\right)\Big|_{(0,0,0)}=\left(0,0,2\right)$$

Then the tangent plane's equations are

$$xF_x(0,0,0)+yF_y(0,0,0)+zF_z(0,0,0)=0\Rightarrow -2z=0\Rightarrow z=0$$
$$xG_x(0,0,0)+yG_y(0,0,0)+zG_z(0,0,0)=0\Rightarrow 2z=0\Rightarrow z=0$$

Hence, the surfaces of $x^2+y^2+(z-1)^2=1$ and $x^2+y^2+(z+1)^2=1$ are tangent to each other at (0, 0, 0).

16. Find the tangent plane to the surface $z=\frac{x^2y}{x^4+2y^2}$ at the point $\left(-1,1,\frac{1}{3}\right)$.

Solution. Tangent plane equation is $z-c=(x-a)z_x(a,b)+(y-b)z_y(a,b)$.

$$z_x=\frac{2xy}{x^4+2y^2}-\frac{4x^5y}{(x^4+2y^2)^2}\Rightarrow z_x(-1,1)=-\frac{2}{9}$$
$$z_y=\frac{x^2}{x^4+2y^2}-\frac{4x^2y^2}{(x^4+2y^2)^2}\Rightarrow z_y(-1,1)=-\frac{1}{9}$$

So, the tangent plane equation is $z=\frac{1}{3}-\frac{2}{9}(x+1)-\frac{1}{9}(y-1)$ or $2x+y+9z=2$.

17. Find equations of the tangent plane at the given point.

(a) $f(x,y) = x^2 - y^2$ at $(-2, 1)$ (b) $g(x,y) = e^{xy}$ at $(2, 0)$

Solution. (a) Let $z = x^2 - y^2 \Rightarrow \begin{cases} z_x = 2x \\ z_y = -2y \end{cases} \Rightarrow \begin{cases} z_x(-2,1) = -4 \\ z_y(-2,1) = -2 \\ z(-2,1) = 3 \end{cases}$.

Tangent plane equation at $(-2,1,3)$: $z - 3 = -4(x+2) - 2(y-1) \Rightarrow 4x + 2y + z = -3$.

(b) $z = e^{xy} \Rightarrow \begin{cases} z_x = ye^{xy} \\ z_y = xe^{xy} \end{cases} \Rightarrow \begin{cases} z_x(2,0) = 0 \\ z_y(2,0) = 2 \\ z(2,0) = 1 \end{cases}$.

Tangent plane equation at $(2,0,1)$: $z - 1 = 0 + 2(y-0) \Rightarrow z = 2y + 1$.

18. Find, if possible, a function $f(x,y)$ for which $f_x(x,y) = e^y$ and $f_y(x,y) = e^x$.

Solution. Assume there is a function $f(x,y)$ such that $f_x(x,y) = e^y$ and $f_y(x,y) = e^x$. Then by using the mixed derivatives theorem, $f_{xy} = f_{yx} \Rightarrow \frac{\partial}{\partial y} e^y = \frac{\partial}{\partial x} e^x \Rightarrow e^y = e^x$, but if we partially differentiate both sides with x, we get $0 = e^x$, a contradiction.

19. Find all second partial derivatives of $f(x,y) = \sqrt{x^2 + 5y^2}$.

Solution. All the first order partial derivatives: $f_x = \frac{x}{\sqrt{x^2+5y^2}}$ and $f_y = \frac{5y}{\sqrt{x^2+5y^2}}$.

All the second order: $f_{xx} = \frac{5y^2}{(x^2+5y^2)^{3/2}}, f_{yy} = \frac{5x^2}{(x^2+5y^2)^{3/2}}, f_{xy} = f_{yx} = -\frac{5xy}{(x^2+5y^2)^{3/2}}$.

20. Let $f(r,\theta) = r^m \cos(m\theta)$ be a function of r and θ, where m is a positive integer.

(a) Find the second order partial derivatives $f_{rr}, f_{r\theta}, f_{\theta\theta}$ and evaluate their respective values at $(r, \theta) = (1, 0)$.

(b) Determine the value of the real number λ so that $f(r,\theta)$ satisfies the differential equation $f_{rr} + \frac{\lambda}{r} f_r + \frac{1}{r^2} f_{\theta\theta} = 0$.

Solution. (a) $f_r = mr^{m-1} \cos m\theta, f_\theta = -mr^m \sin m\theta$

$\Rightarrow f_{rr} = m(m-1)r^{m-2} \cos m\theta, f_{r\theta} = -m^2 r^{m-1} \sin m\theta, f_{\theta\theta} = -m^2 r^m \cos m\theta$

$\Rightarrow f_{rr}(1,0) = m(m-1), f_{r\theta}(1,0) = 0, f_{\theta\theta}(1,0) = -m^2$

(b) $f_{rr} + \frac{\lambda}{r} f_r + \frac{1}{r^2} f_{\theta\theta} = 0 \Rightarrow m(m-1)r^{m-2} \cos m\theta + \lambda m r^{m-2} \cos m\theta - m^2 r^{m-2} \cos m\theta = 0$

$\Rightarrow (\lambda - 1) m r^{m-2} \cos m\theta = 0 \Rightarrow \lambda = 1$

$\lambda = 1$ guarantees that $f_{rr} + \frac{\lambda}{r} f_r + \frac{1}{r^2} f_{\theta\theta} = 0$ for any r and θ (and m is positive integer).

Past years test/final exam questions

21. Let $f(x,y)=\begin{cases}\frac{2y^2}{2x-y}, & x\neq y\\ 0, & x=y\end{cases}$. Show by definition, that $f_x(0,0)$ and $f_y(0,0)$ exist, and compute their value.

Solution. $f_x(0,0)=\lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=\lim_{h\to 0}\frac{0/h-0}{h}=0$; $f_y(0,0)=\lim_{h\to 0}\frac{f(0,h)-f(0,0)}{h}=\lim_{h\to 0}\frac{\frac{-2h^2}{h}-0}{h}=-2$.

Hence, $f_x(0,0)$ and $f_y(0,0)$ exist, and $f_x(0,0)=0$ and $f_y(0,0)=-2$.

22. Find the tangent's plane equation for function $h(x,y)=\ln\left(1+\frac{x}{y}\right), x,y>0$ passes through the point with $(x,y)=(2,1)$.

Solution. $h(2,1)=\ln 3$; $h_x=\frac{1/y}{1+x/y}\Rightarrow h_x(2,1)=\frac{1}{3}$; $h_y=\frac{-x/y^2}{1+x/y}\Rightarrow h_y(2,1)=-\frac{2}{3}$.

Tangent plane: $z=\ln 3+\frac{1}{3}(x-2)-\frac{2}{3}(y-1)$.

23. Let $f(x,y,z)=x^2+xy+y^2-z$.

(a) Find ∇f, the gradient of f.

(b) Find the equation for the tangent plane for $f(x,y,z)=0$ at $(1,0,1)$.

Solution. (a) $\nabla f=(2x+y,x+2y,-1)$ (b) $\nabla f(1,0,1)=(2,1,-1)$.

Tangent plane: $2(x-1)+(y-0)-(z-1)=0\Rightarrow 2x+y-z=1$.

24. Let $f:\mathbb{R}^2\to\mathbb{R}$ be a function defined as $f(x,y)=\begin{cases}\frac{xy}{x^2+y^2} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0)\end{cases}$. Find $\left.\frac{\partial f}{\partial x}\right|_{(0,0)}$ using the definition of partial derivative. Guess $\left.\frac{\partial f}{\partial y}\right|_{(0,0)}$ without computation.

Solution. $\left.\frac{\partial f}{\partial x}\right|_{(0,0)}=\lim_{h\to 0}\frac{f(h,0)-f(0,0)}{h}=\lim_{h\to 0}\frac{0}{h}=0$ and $\left.\frac{\partial f}{\partial y}\right|_{(0,0)}=0$.

25. Let $k\in\mathbb{R}$. Prove that the tangent plane to the graph of the function $f:\mathbb{R}^2\to\mathbb{R}$ given by $f(x,y)=\sin(xy)$ at the point $(0,k)$ has equation $z=kx$.

Solution. Let $g(x,y,z)=\sin(xy)-z$ so that $g=0$ at $(0,k,0)$.

$\nabla g=(y\cos(xy),x\cos(xy),-1)\Rightarrow\nabla g(0,k,0)=(k,0,-1)$. So the tangent plane: $z=kx$.

Alternative. The tangent plane is given by $z=f(0,k)+xf_x(0,k)+(y-k)f_y(0,k)$. Now

$$\begin{cases}f(x,y)=\sin(xy)\\ f(0,k)=0\end{cases}\Rightarrow\begin{cases}f_x(x,y)=y\cos(xy)\Rightarrow f_x(0,k)=k\\ f_y(x,y)=x\cos(xy)\Rightarrow f_y(0,k)=0\end{cases}$$

Hence the tangent plane is given by $z=kx$.